Collection Framework introduction

In coding we are dealing with code that is stored in object inside (object oriented Programming). Taki hum alag-alag QUERY'S kr ske.

DATASTRUCTURES→

- The need of **DataStructure** is for storing **data** efficiently and also to perform **queries** on the stored data.
- If there is a Arry of numbers, [1, 23, 5, 45, 7,87,76,43,5,0]--> this **datastructure** is not organized.
- And if we have **query** to get the biggest number→therfore, in **time complexity** muj laga **Order of (N)** time. Or ye query mushkil ho gi.
- Therefore, **organized datastructure** for the above **query** is→ array in ascending order, and we will get the answer in **constant time**→ order of (1).

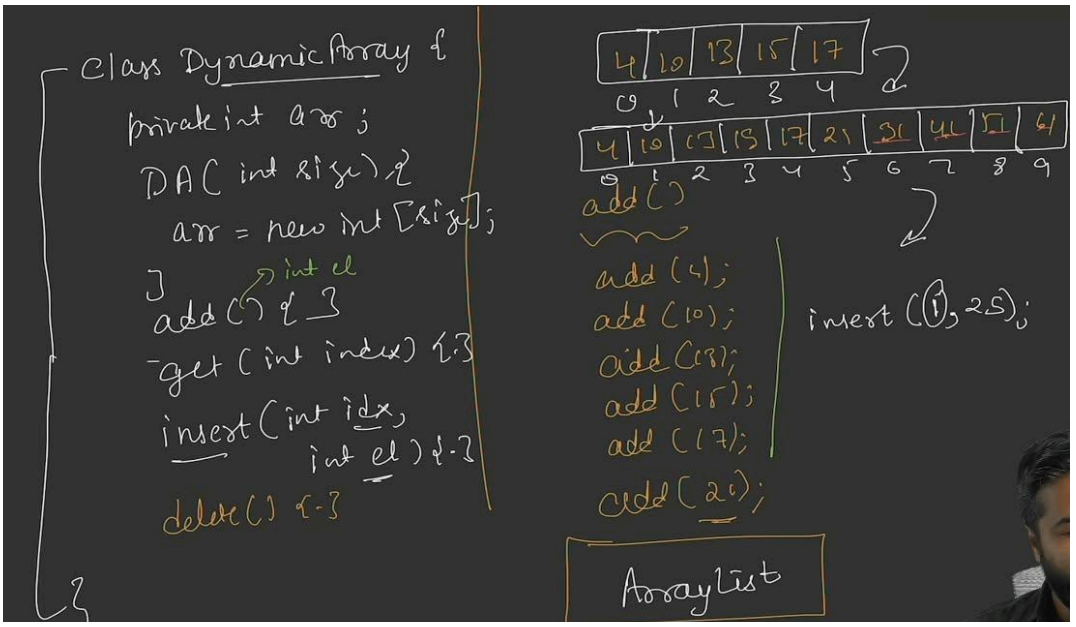
Therefore, kisi query ko easily access krne ke liye organized datastructures ka framework ko collection framework khte ha.

- Java provides support in general all datastructers, by giving classes.

Data Structures Types

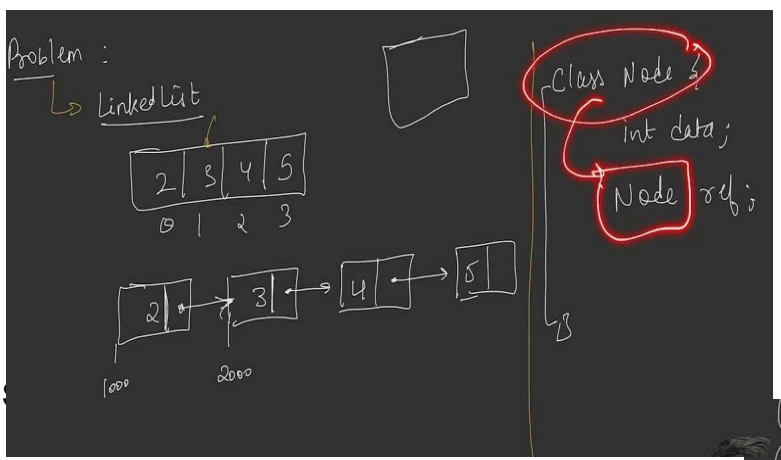
Dynamic Array DataStructure:

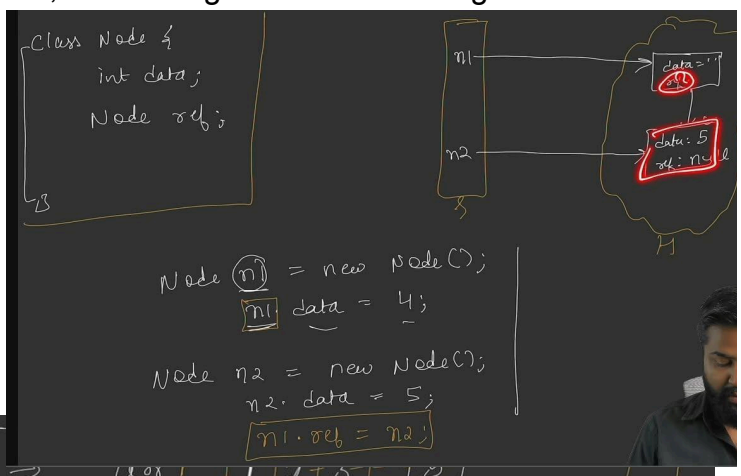
- isme Array ka koi fixed size nahi ha, size demand ke hisab se extend hota ha.
- Dynamic Array ke saare kaam ArrayList me ho skte ha jo java support krta ha.



Linked list DataStructure:

- It dosen't store data in the **contiguous memory address** like **Array.**
- It stores the reference of the next value of same node, iska storage kbhi full nahi hoga.





- Duplicates are not allowed in this dataStructure.
- Tell in constant-time that element exists or not.
- This is possible by the dataStructure here→
- This is with particular range of array.
- For without range Array it is done with modulus method.

Stack & **Queues** DataStructure**:**

Stack:

- It means ki jis order me **data** Array me dala ha, uske **opposite** order me data ko nikalna ha.
- Iske entry or exit points **same** hote ha.
- It is also known as **LIFO→ Last in, First out,** in programming terms.
- Isme hume Array ke koi bhi access ni milte ha, taki hum use bich se access na kr ske.
- Only add(); & remove(); methods should be given.

Queues:

- It means ki jis order me **data** Arrray me dala ha, us hi **order** me data ko nikalna ha.
- It is also known as **FIFO→ First in, First out,** in programming terms.
- Isme Arrays ke entry & exit points alag-alag hote ha.
- Entry samne se or exit piche se hoti ha.
- Only add(); & remove(); methods should be given, same but with different style.

Maps DataStructure:

- It is as similar as Set DataStructure,
- .bs isme values ke sath proper keys hoti hai, hum pairs me store krte ha.
- Example→ Rohit, 101 / Aditya, 102.
- No duplicate values and keys are not allowed.

Tree DataStructure:

- **BST: Binary Search Tree→**
- 1node only have 2 childs.
- Left child is smaller then node.
- Right child is greater then node.

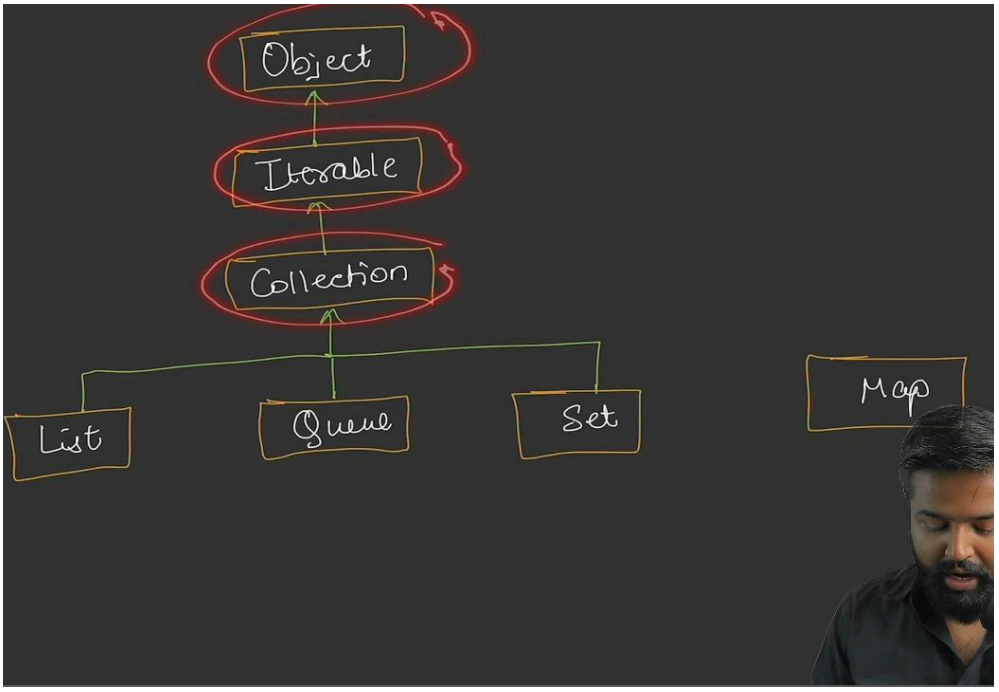
Java has given 2 classes for BST tree DataStructure.

- Class treeSet
- Class TreeMap

Note:

- All DataStructure are in **hierarchy**.
- **ArrayList** and **LinkList** both are child of **interface List.**
- Now, **List, Set, queues, stack,** etc all these interfaces are the **child** of the **parent interface**

COLLECTION.



- **Map is hierarchy ki bahar ha kuki ye key or value ke sath deal krta ha.**
- **Baki sabhi dataStructures values ke sath deal krte ha.**